

Geomodelling - Geostatistics for Natural Resource Evaluation

Report Variography

1 Task description

Insert here the task.

2 Empirical variogram

a rough estimate of nugget, sill and range, plot

3 Variogrammodelling

Which variogrammodel would fit? Simple model or nested? Fit the model. Give the parameter of the fitted model, draw it.

4 Anisotropy

Check the data for anisotropy, variogram map. Can you fit a directional variogram? Do you have enough data? If possible fit a directional variogram model and give the parameter of the model (main angles, directional sill, range and nugget), draw it.

5 Results

Which variogrammodel you will use for the further analysis? Why?

Checklist – During the variogram analysis you should do the following steps and try to answer the questions in your report:

- look at the variogram cloud, does anything looks strange, up to which distances we have many pairs, are there outliers/ outlier bands, extrem points
- empirical variogram - try to guess up to which distance you can interpret it
- which variogrammodel would fit? Does a simple model fit or do we need a nested model?
- give a rough estimate of nugget, sill and range
- fit a variogram model
- check the fit
- do you have enough points (visual check / 30 - 50 pairs in the groups)?
- draw your variogram model
- if it doesn't fit, decide on directional range and sill
- check for directional dependency - draw a variogram map
- if it looks like directional dependency find main angles and draw directional variograms
- do you have enough points (30 - 50 pairs in the groups)?
- prepare a directional variogram model, fit it, check it